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The Healing Components of Rice Bran

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INTRODUCTION

Rice is one of the most important staple foods for a large part of the human population. Global rice production is 645 million tonnes and this huge amount of production results in large amount of rice's by-products (Kubglomsong & Theerakulkait, 2014; Al-Okbi *et al*., 2014). Brown rice is produced from dehusking of rice grain and it is covered by a layer of bran. Rice bran is obtained by debranning or during polishing process; it constitutes 5-10% of brown rice (Kubglomsong & Theerakulkait, 2014). Rice bran is part of the hard outer layer of the grain that contains pericarp, aleurone and subaleurone fractions (Figure 16.1). With some proportions of endosperm and germ, rice bran is often produced as a by-product of rice milling in the production of refined rice grains. It is estimated that the annual world production of rice bran amounts to 76 million tonnes (Kahlon, 2009).

Rice milling process consists of two basic procedures; one of them is removal of the husk to produce brown rice and the other one is the removal of the bran layer from brown rice to produce polished (or white) rice. The milling process also removes the germ and a portion of the endosperm as broken kernels and powdery materials (Champagne *et al*., 2004). Hence, the output of a milling process comprises one primary product which is milled rice, and several types of by-product which are the husk, germ, bran layer and broken kernels (Lloyd *et al*., 2000).

Malaysian rice bran is considered a waste and underutilised. It has been used as an ingredient in the formulation of animal feeds. However, it is a good source of food ingredient for human consumption. In spite of being an excellent source of nutrient, raw rice bran is not suitable for human consumption mainly due to the rancidity problem. Freshly milled rice bran has short shelf-life owing to the decomposition of lipids (triacylglycerols) into free fatty acids (FFA) by lipases, making it unsuitable for fresh consumption (Barnes & Galliard, 1991).

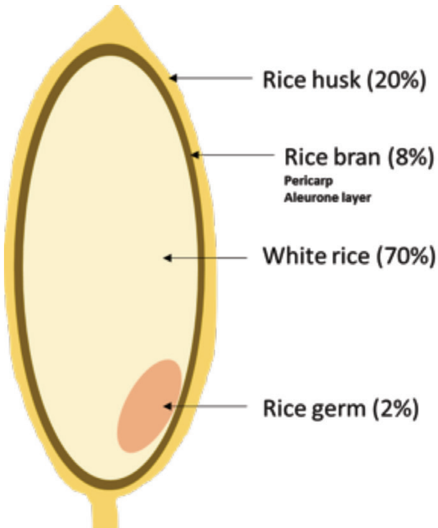

Figure 16.1 The structure of a rice grain

A major obstacle in using rice bran is that it rapidly deteriorates; thus, it is necessary to find a good way for preventing the rapid deterioration of rice bran and to ensure its quality for further processing. Rice bran must be stabilised immediately upon production due to the presence of lipase, an enzyme that rapidly hydrolyses oil to free fatty acid (FFA) and glycerol, therefore reducing the quality of rice bran (Enochian *et al*., 1981). Stabilisation of rice bran by microwave heating is a popular and effective method that has been recently studied (Ramezanzadeh *et al*., 2000). Besides, extrusion, ohmic heating, dry heat treatment, gamma radiation, parboiling and roasting are also used for stabilisation of rice bran (da Silva *et al*., 2006; Tuncel *et al*., 2014).

As mentioned earlier, rice bran (which includes the germ) is accounting for approximately 10% of the weight of rice grain (Jan *et al*., 2010; Kahlon, 2009). Comprising of just 10% of total weight, rice bran accounts for most of the nutrients found in each rice kernel. When the bran is removed from rice grains, the grains lose a portion of their nutritional value. All types of rice grain have rice bran, bran layer and rice germ ranged between 17.59-21.75%, 14.63-17.14% and 2.81-3.19% respectively (Moongngarm *et al*., 2012). This chapter will discuss on food and nutritional components, health benefits and potential applications of rice bran.

FOOD AND NUTRITIONAL COMPONENTS

The chemical and nutritional qualities of rice grain vary considerably and this may be attributed to genetic factors, environmental influences, used of fertiliser, degree of milling and storage conditions. As a result, rice bran obtained from different rice varieties exhibits different chemical compositions and nutritional qualities (Houston, 1972). Rice bran provides dietary fibre-rich total carbohydrates, protein, B vitamins and minerals as well as the essential fatty acid in the rice bran oil. Besides the essential nutrients, rice bran also contains a range of phytochemicals which may provide some of the health benefits seen among the populations consuming plant-based diets. However, rice bran contains some anti-nutrients.

Macronutrients

Macronutrients components of the bran that is stabilised by autoclaving or parboiling at 4% and 8% milling degrees are presented in Table 16.1. Different milling degrees affect the chemical and nutritional qualities of rice bran. Rosniyana *et al.* (2005) reported that rice bran produced at 4% milling degree was the most nutritious compared with the brans produced at other milling degrees. Rice bran is predominantly composed of carbohydrate, fat and protein. The results also showed that carbohydrate content of rice bran varied between 25.91% and 47.14%. In addition, the fat (19.4-30.45%) and protein (13.6-19.5%) contents were markedly higher in the bran obtained from rice that milled at 4% milling degree. The percentages of ash were also found in the range of 7.7-11.1% and moisture content of 4.96-5.6%. Other components were dietary fibre (20-23.5 g/100 g) and soluble fibre (1.9-2.1 g/100 g).

Rice bran has high percentage of fat (10-23%) by weight as compared with the other brans derived from grains, as well as 4.2% unsaponifiable matter (Kahlon & Woodruff, 2003; Sugano *et al*., 1998; Sharif *et al*., 2014), negligible amount of water-soluble β-glucans and larger amount of insoluble dietary fibre (Rukmini & Raghuram, 1991). Rice bran oil has a higher percentage of polyunsaturated fatty acids (PUFA) than its saturated counterparts. It contains a relatively large proportion of the important unsaturated fatty acids such as oleic acid (52.1% in the cold 80% methanolic extract and 34.8% in hot hexane extract) and linoleic acid (22.1% in the cold and 27.1% in the hot extracts). Major saturated fatty acid presence in rice bran oil is palmitic acid which is 16.2% in the cold and 14.9% in the hot extracts (Oluremi *et al*., 2013). The saponifiable lipids in rice bran oil include 68-71% triglycerides, 2-3% diglycerides, 5-6% monoglycerides, 2-3% FFAs, 2-3% waxes, 5-7% glycolipids and 3-4% phospholipids (McCaskill & Zhang, 1999). In addition to the nutritional components of rice bran oil, other properties such as good stability, appealing flavour and extending oil frying-life, make rice bran oil as an alternate to bakery shortening (Liang *et al*., 2014).

Rice bran, its oil and hull contain a large number of bioactive compounds, with pigmented bran has many types of bioactive compound than the white bran. For instance, rice bran is the best source of phytosterols based on raw material (Friedman, 2013). The class of sterols, known as phytosterols, is found mainly in plant cell walls and membranes. Phytosterols including stanols cannot be synthesised by human body; all plant sterols and stanols in the human body are obtained from plant-based diet. Sitosterol, campesterol and stigmasterol are the major phytosterols in the lipid extracts of rice bran which have hypocholesterolemic effect (Jiang & Wang, 2005; de Jong *et al*., 2003). Rice bran is reported as the most lipid-rich material, with 22.2% of total lipid extracted based on a dry weight basis. In terms of raw material, a high phytosterol content (4.5 mg/g bran) is also obtained from rice bran (Jiang & Wang, 2005).

Table 16.1 Macronutrient components of stabilised rice bran with different degrees of milling

Nutrient (%)	Rice bran (Autoclaved)		Rice bran (Parboiled)	
	4% milling degree	8% milling degree	4% milling degree	8% milling degree
Moisture	5.13±0.05[a]	4.96±0.15[a]	5.02±0.20[a]	5.60±0.10[a]
Ash	7.68±0.01[b]	7.65±0.05[b]	10.84±0.75[a]	11.08±0.75[a]
Fat	19.50±0.01[c]	19.40±0.75[c]	30.45±0.25[a]	25.50±0.10[b]
Protein	14.69±0.25[a]	13.60±0.55[b]	15.10±0.35[a]	13.60±0.15[b]
Crude fibre	7.80±0.15[b]	7.25±0.75[b]	12.68±0.25[a]	9.80±0.10[c]
Carbohydrate	45.20±0.75[b]	47.14±0.25[a]	25.91±0.75[d]	34.42±0.75[c]
Dietary fibre (g/100 g)	23.3±0.05[a]	22.4±0.75[a]	23.5±0.15[a]	20±0.25[b]
Soluble fibre (g/100 g)	2.1±0.75[a]	1.9±0.05[a]	2.2±0.00[a]	2.0±0.75[a]

Mean values in the same row with different superscript lowercase letters ([a-d]) are significantly different at $p<0.05$

Source: Rosniyana *et al*. (2009)

Micronutrients

Consumption of rice bran contributes to the increase intake of vitamins (Table 16.2) and minerals (Table 16.3) although the micronutrients content is depending on the stabilisation techniques and milling degrees (Rosniyana *et al*., 2009). Rice bran is rich in niacin (30-37 mg/100 g), followed by pyridoxine (3.4-4.2 mg/100 g), thiamine (2.2-2.4 mg/100 g) and riboflavin (0.25-0.30 mg/100 g). It is also rich in minerals such as phosphorus and potassium where the amounts in rice bran ranged from 1710 to 1830 mg/100 g and from 1170 to 1350 mg/100 g respectively. Other minerals are calcium, sodium and magnesium which markedly higher in autoclaved rice bran in comparison to parboiled rice bran. In contrast, amount of iron is considerably higher in parboiled rice bran than autoclaved rice bran (Table 16.3).

Table 16.2 Vitamins component of stabilised rice bran with different degrees of milling

Nutrient (mg/100 g)	Rice bran (Autoclaved)		Rice bran (Parboiled)	
	4% milling degree	**8% milling degree**	**4% milling degree**	**8% milling degree**
Thiamin	2.4±0.25[a]	2.3±0.05[a]	2.4±0.75[a]	2.2±0.05[a]
Riboflavin	0.25±0.05[c]	0.27±0.75[b]	0.3±0.75[a]	0.29±0.05[a]
Niacin	30.0±0.25[c]	36.0±0.25[b]	37.0±0.75[a]	37.0±0.05[a]
Pyridoxine	3.4±0.75[d]	3.7±0.05[c]	4.2±0.15[a]	4.0±0.75[b]

Mean values in the same row with different superscript lowercase letters ([a-d]) are significantly different at $p<0.05$

Source: Rosniyana *et al*. (2009)

Table 16.3 Minerals component of stabilised rice bran with different degrees of milling

Nutrient (mg/100 g)	Rice bran (Autoclaved)		Rice bran (Parboiled)	
	4% milling degree	**8% milling degree**	**4% milling degree**	**8% milling degree**
Calcium	54±0.05[b]	55±0.15[b]	62±0.70[a]	61±0.75[a]
Magnesium	823±0.25[d]	845±0.15[c]	952±0.25[a]	940±0.25[b]
Sodium	24±0.25[b]	25±0.15[b]	28±0.05[a]	28±0.75[a]
Potassium	1170±0.25[c]	1200±0.75[b]	1350±0.75[a]	1340±0.25[a]
Phosphorus	1180±0.05[b]	1830±0.25[a]	1710±0.25[b]	1800±0.15[a]
Iron	14±0.75[d]	19±0.75[c]	23±0.15[b]	30±0.25[b]

Mean values in the same row with different superscript lowercase letters ([a-d]) are significantly different at $p<0.05$

Source: Rosniyana *et al*. (2009)

Phytochemicals

Phytochemical is the non-nutritive chemicals of plants. For instance, rice bran contains a range of phytochemicals such as phenolic acids, flavonoids, anthocyanins and steroidal compounds (Table 16.4). These phytochemicals have health-promoting effects that are often referred as bioactive substanes (Friedman, 2013). Rice bran is also rich in phytic acid and γ-oryzanol. In addition, α-tocopherol and γ-tocopherol are the highest in rice germ (Moongngarm *et al*., 2012). The levels of total tocopherols and total tocotrienols in rice bran samples ranged from 319.67 to 443.73 μg/g rice bran. In the past, γ-tocotrienol has been used as a natural antioxidant for food products. Also, α-tocopherol has high bioavailability and bioactivity for human health. The concentrations of γ-oryzanol in rice bran ranged from 3861.93 to 5911.12 μg/g rice bran (Min *et al*., 2011).

Attention begins to be focused on the components of rice bran oil. Crude rice bran oil is rich in bioactive compounds such as γ-oryzanol, phytosterols, tocopherols and tocotrienols as well as possible hypocholesterolaemic agents (Friedman, 2013; Sugano & Tsuji, 1997). Anthocyanins can be found in different varieties of rice bran, particularly the highest in purple rice bran which is the most abundant and economical natural source of anthocyanin that can be applied as a natural colourant and being used for health-promoting ingredient as nutraceutical and functional food (Min *et al*., 2011). Anthocyanin is one of the subclasses of flavonoid and it is a natural colourant. As an example, red and purple pigments in most fruit, vegetables and cereal grains including pigmented rice in particular purple rice bran are the natural colourants. Anthocyanins also have been reported to possess strong antioxidant capacity and many health benefits (Mazza, 2007).

Table 16.4 Bioactive compounds in rice bran

Phenolic acids	Anthocyanins, flavonoids	Steroidal compounds
Caffeic acid	Anthocyanin monomers, dimers and polymers	Acylatedsteryl glucoside
Coumaric acid	Apigenin	Campesterol ferulate
Catechin	Cyanidin glucoside	γ-Oryzanol
Ferulic acid	Epicatechin	β-Sitosterol
Gallic acid	Hesperetin	Tocopherol
Hydroxybenzoic acid	Luteolin	Tocotrienol
Methoxycinnamic acid		
Vanillic acid		
Sinaptic acid		
Syringic acid		

Source: Friedman (2013)

Another component of rice bran is inositol, which is the vitamin-like sugar alcohol. Deficiency in inositol is associated with alopecia (hair loss) and abnormal development (Jariwalla, 2000). Inositol exists in three forms, as free inositol, as part of the component of phospholipids and as inositol phosphates or phytate.

Anti-Nutrients

In most rice grain, phytic acid is concentrated in the bran layer, particularly aleurone layer, and to a lesser extent, the germ (Lasztity & Lasztity, 1990). Most antinutritional factors found in rice bran (trypsin inhibitor, lectins and phytic acid or also known as phytate) are inactivated by heat and form insoluble complexes (Oatway *et al*., 2001). Garcia *et al*. (2012) reported that the content of phytic phosphorus was consequently higher in raw rice bran compared to that of the stabilised bran. Literature also showed that phytic acid has the structure that capable to chelate minerals such as calcium, zinc and iron. It has antioxidative effect (Zhou *et al*., 2003) and combines with protein and starch result in reduced bioavailability of these essential nutrients.

Haemagglutinin, a rice bran's lectin, is higher in raw bran, and the lectin activity is lower in stabilised bran (Khan *et al*., 2009). Haemagglutinin-lectin is capable of binding to specific carbohydrate receptor sites in the intestinal wall, thereby lowering nutrient absorption (Goldstein, 1980). Trypsin inhibitors are responsible for impairing trypsin enzyme activity in the digestive tract (Bahnassey *et al*., 1986). Khan *et al*. (2009) also reported the highest activity observed in raw rice bran (8.44 activity/mg), while stabilised rice brans showed no detectable activity of trypsin inhibitors.

HEALTH BENEFITS

Antioxidant Properties

Rice bran contains a unique complex of naturally occurring antioxidant compounds (Moldenhauer *et al*., 1998). Antioxidant compounds in rice bran (Canan *et al*., 2012) or rice bran oil have significant health benefits. This observation is supported by a previous study that rice bran is an important source of phytochemicals that has antioxidative and disease-preventing properties, particularly cancer (Devi & Arumughan, 2007). Another study also reported antioxidant capacity of rice bran that assessed using DPPH radical scavenging activity (Moongngarm *et al*., 2012).

Rice bran is an excellent source of antioxidants such as polyphenols, tocopherols, tocotrienols and γ-oryzanol which help in preventing oxidative damage of body tissues and DNA (Tuncel *et al*., 2014). The presence of tocopherols and tocotrienols in rice bran extract and the synergistic effect of these compounds are reported as the reasons for the strong antioxidant activity (Arab *et al*., 2011). Norhaizan *et al*. (2011) revealed that phytic acid extracted from rice bran had antioxidant capacity that tested using ferric thiocyanate capacity (FTC), thiobarbituric acid (TBA), ferric reducing antioxidant power (FRAP) and β-carotene bleaching methods. They also reported a possible mechanism of phytic acid as an antioxidant for preventing the generation of reactive oxygen species with its metal-binding properties (Norhaizan *et al*., 2011).

Anticancer Effect

Apart from being a good source of essential nutrients such as vitamins, minerals and fibre, rice bran also has many bioactive components which are currently drawing a lot of attention for prevention and treatment of many types of cancer. For instance, rice bran phytonutrients are related to the prevention as well as slowing the progression of certain cancers such as breast, colon, lung, liver and other cancers (Hudson *et al*., 2000).

Inositol in rice bran is a chemopreventive agent with low toxicity to normal cells and has the ability to inhibit cancer development in various organs; for examples, mammary gland, colon and lung (Jariwalla, 2000; Wattenberg, 1998). Other studies also showed chemopreventive efficacy of inositol from rice bran using *in vivo* models such as liver carcinogenesis in mice (Nishino *et al*., 1998) and colon carcinogenesis in rats (Shamsuddin & Ullah, 1989).

While phytic acid is considered as an anti-nutritional component in cereals, seeds and beans, some people might concern that a high intake of phytic acid in food is thought to reduce the bioavailability of dietary minerals. However, previous studies have shown that phytic acid does not appear to affect mineral balances unless a large quantity of inositol hexaphosphate is consumed (Norhaizan & Norfaizadatul, 2009; Sandström *et al*., 2000). Chemopreventive efficacy of phytic acid from rice bran has been performed based on *in vitro* and *in vivo* colon cancer models (Nurul-Husna *et al*., 2010; Shafie *et al*., 2013a,b). For instance, phytic acid in rice bran inhibits cell cycle progression and further induces early apoptosis in HepG2 liver cancer cell lines (Norazalina *et al*., 2013). Phytic acid given in drinking water also inhibits colon tumour growth in Sprague-Dawley rats which is associated with anti-proliferative and pro-apoptotic effects of phytic acid *in vivo* (Shafie *et al*., 2013b). Another study revealed the administration of 0.2% phytic acid in rice bran inhibited the formation of aberrant crypt foci and the incidence of tumour growth (Shafie *et al*., 2013b; Norazalina *et al*., 2010). All these findings strengthen the fact that phytic acid can modulate various molecular pathways underlying the multistep process of carcinogenesis (Matejuk & Shamsuddin, 2010).

In vitro studies showed that rice bran inhibited the growth of various types of cancer cell lines including endometrial (Fan *et al*., 2000), leukaemia (Ghoneum & Gollapudi, 2003), breast (Gollapudi & Ghoneum, 2008), colon and liver (Kannan *et al*., 2010), melanoma (Chung *et al*., 2009), pancreatic (Kunnumakkara *et al*., 2010) and ovarian (Norhaizan *et al*., 2011). *In vivo* studies involving laboratory mice revealed that the several bran preparations inhibited breast tumours (Cai *et al*., 2004), urinary bladder tumours (Kuno *et al*., 2006), Ehrlich carcinoma tumours (Badr El-Din *et al*., 2008), colorectal carcinogenesis (Phutthaphadoong *et al*., 2009) and head and neck carcinomas (Fujiwara *et al*., 2011). Other studies using rats showed that rice bran and its bioactive compounds inhibited carcinogenesis of liver (Katayama *et al*., 2003), oral (Long *et al*., 2007), gastric (Tomita *et al*., 2008) and colon (Norazalina *et al*., 2013). Friedman (2013) reported that tumour mass was 35% lower after administration of black rice bran and 19% lower after administration of brown rice bran in the transplanted colon tumour of mice via inhibition of tumour biomarkers including expressions of cyclooxygenase-2 (COX-2), 5-lipoxygenase (5-LOX) and vascular endothelial growth factor as well as inhibition of neo-angiogenesis inside the tumours.

A clinical study using a commercial product of arabinoxylan rice bran (MGN-3-BioBran) showed the enhanced effect of interventional therapy for the treatment of hepatocellular carcinoma patients (Bang *et al*., 2010). The results showed that there was a lower recurrence of the disease (31.6%) than control (47.7%) and a higher survival of 35% in the second year than the control (6.7%). This finding indicates the potential of rice bran in treatment of liver cancer. Furthermore, the MGN-3 arabinoxylan rice bran shows increased of natural killer cell activity and dendritic cells in peripheral blood as well as an augmented increased of T-helper cells in multiple myeloma patients (Cholujova *et al*., 2013).

Anti-Hypercholesterolaemic Effect

It is postulated that tocotrienols lower cholesterol through inhibition of 3-hydroxy-3-methyl-glutaryl-coenzyme A (HMG-CoA) reductase, the rate-limiting enzyme in endogenous cholesterol synthesis (Kerckhoffs *et al*., 2002). A study reported that the tocotrienol-rich fraction of rice bran when taken in combination with an American Heart Association Step 1 diet, the fraction reduced serum total cholestrol and LDL-cholesterol concentrations in the hypercholesterolaemic subjects (Qureshi *et al*., 1997; 2002). The total cholesterol and LDL-cholesterol-lowering effects of rice bran oil have also been demonstrated based on a human model (Most *et al*., 2005). The results showed that combination of fatty acids in rice bran oil with a control oil blend had effect on serum cholesterol concentration which is due to the unsaponifiable compounds present in it but not due to its fatty acids.

Rice bran oil that contains these compounds is an important functional food with cardiovascular health benefit (Most *et al*., 2005). Literature showed that rice bran oil, possibly due to the unsaponifiable fraction or its fatty acid content, decreased cholesterol levels in hamsters, rats, humans and non-human primates (Sharma & Rukmini, 1986; Seetharamaiah & Chandrasekhara, 1989; Nicolosi *et al*., 1991; Kahlon *et al*., 1992; Purushothama *et al*., 1995).

Rice bran oil, but not the fibre, lowers blood lipids in men and women with borderline high total cholesterol (Most *et al*., 2005). Therefore, it is believed that rice bran reduces cholesterol by a mechanism different from that of oat bran. Other studies also demonstrated that decreased in cholesterol level were found in hypercholesterolaemic subjects who replaced their usual cooking oils with rice bran oil (Raghuram *et al*., 1989) and in middle-aged and elderly subjects consuming a low-fat diet containing rice bran oil (Lichtenstein *et al*., 1994).

Hu and Yu (2013) reported the scavenging effects of rice bran fibre on cholesterol and bile acids *in vitro*. The binding was high with hemicellulose and poor binding with cellulose, insoluble dietary fibre and lignin in the rice bran fractions. Various rice bran formulations also shown to reduce cholesterol levels in chicken (Newman *et al*., 1992), hamsters (Kahlon *et al*., 1992; Wilson *et al*., 2007; Rong *et al*., 1997), mice (Qureshi *et al*., 2001a) and swine (Qureshi *et al*., 2001b). The bioactive fractions and components of rice bran are clinically proven to reduce plasma lipids in hypercholesterolaemic individuals but apparently not in individuals with normal plasma lipid profile (Rondanelli *et al.*, 2011).

Anti-Hyperlipidaemic Effect

A study by Jariwalla (1998) demonstrated that phytic acid in rice bran lowered lipid cholesterol and triacylglycerol in serum indicating a role in inhibiting cardiovascular diseases. Hyperlipidaemia is defined as a high level of total cholesterol and triacylglycerol in blood, and it is associated with cardiovascular diseases. Phytic acid-supplemented diet is shown to reduce serum cholesterol and triacylglycerol levels in the rats fed with high-cholesterol diet and to down-regulate hypercholesterolaemia marker including zinc/copper ratio. This lipid-lowering agent is observed with no toxic effect in rats (Jariwalla *et al*., 1990).

Another study showed the effect of a combination of inositol and phytic acid in reducing hepatic lipids and triacylglycerol from administration of sucrose via inhibition of hepatic enzymes involved in lipogenesis rather than inhibiting the intestinal enzymes (Katayama, 1998). Phytic acid is also shown to inhibit platelet aggregation (Vucenik *et al.*, 1998), to enhance inflammatory response (Eggleton, 1998) and to inhibit calcification in aorta and lipid peroxidation in kidney which are associated with prevention of cardiovascular diseases (Jariwalla *et al*., 1990).

Anti-Diabetic Effect

Type 2 diabetes is a type of human disease caused by under production of insulin by the pancreas and impaired insulin resistance, leading to hyperglycaemia that results in impaired entry of glucose into the cells, thus hinder glucose utilisation (Yang *et al*., 2012). According to a human study, consumption of stabilised rice bran fractions by the insulin-dependent diabetic subjects resulted in a reduction of glycosylated haemoglobin levels which is a biomarker of diabetes. The results also showed about 25% of the patients had a significant reduction in hyperglycaemia and decreased in daily injection of insulin and hypoglycaemic drugs (Qureshi *et al*., 2002). Another study also reported that stabilised rice bran had significantly lower serum glucose, glycated haemoglobin and lipoprotein as well as increased adiponectin level, suggesting that rice bran might represent a functional nutrient to ameliorate glycaemic and lipid anomalies in type 2 diabetes' patients (Cheng *et al*., 2010).

On the other hand, previous *in vivo* studies demonstrated anti-diabetic effect of rice bran in mice (Kim *et al*., 2010) and rats (Posuwan *et al*., 2013; Kaup *et al.*, 2013; Siddiqui *et al*., 2010). Fermented rice bran also strongly up regulated expression of adiponectin and induced insulin resistance by neutralising free radicals in adipocytes (Kim & Han, 2011; Kim *et al*., 2011). Moreover, γ-oryzanol in rice bran stimulates secretion of insulin by the pancreas (Kozuka *et al*., 2012).

Anti-Allergic Effect

An allergic reaction takes place when the immune system reacts to the exposure of a variety of substances (peanut proteins, soy proteins, pollen and spores) called allergens. Black rice bran from the LK1-3-6-12-1-1 cultivar shows inhibitory effects of histamine and β-hexosaminidinase in the *in vitro* allergic reactions mediated by stimulated basophilic leukaemia and rat's peritoneal mast cells (Nam *et al*., 2005). MGN-3 BioBran, the commercial rice bran previously shown to be a potent biological response modifier activates natural killer cells, T cells and monocytes. It also activates human monocyte dendritic cells *in vitro*, suggesting its for possible use in a dendritic cell-based vaccine against infections (Ghoneum & Agrawal, 2011) and as a potent enhancer of dendritic cell maturation (Cholujova *et al*., 2009). An *in vivo* study discovered the effect of cycloartenylferulate from rice bran oil-derived γ-oryzanol on allergic reaction in the dorsal skin of rats, where cycloartenyl ferulate captured dimmunoglobulin E (IgE) and prevented it from binding to the allergy-related signalling receptor FceR1 as well as attenuated mast cell degranulation (Oka *et al.*, 2010).

Anti-Inflammatory Effect

A study showed that a series of feruloyl esters of triterpene alcohols isolated from a methanolic extract of edible rice bran exhibited anti-inflammatory activity against 12-O-tetradecaolylphorbol-13-actate-induced inflammation in mice (Akihisa *et al*., 2000). In addition to this, acid esters of oligosaccharides from rice bran suppressed the inflammatory mediators (Fang *et al*., 2012), indicating its potential as an anti-inflammatory agent. The hydrolysed rice bran is also known to prevent common cold syndrome in elderly individuals via its immunomodulatory action (Maeda *et al*., 2004) and the cycloartenyl ferulate extracted from rice bran oil reduces lipopolysaccharide-induced nitric oxide production and mRNA expression of NO synthase and COX-2 as well as up regulate superoxide dismutase activity. The finding suggests its value for the therapy of inflammatory diseases (Nagasaka *et al*., 2007).

A previous study reported that the stabilised rice bran extract had inhibitory effect on the pro inflammatory enzymes, COX-1, COX-2 and 5-LOX (Roschek Jr *et al*., 2009). Another *in vivo* study also reported that the prebiotics from rice bran ameliorated inflammation in murinecolitis by regulating immune cell differentiation, suppressed the growth of *Clostridium* and increased short-chain fatty acid content in colitis (Komiyama *et al*., 2011). Besides the findings, a component of the bran by-product of sake brewing rice, cycloartenyl trans-ferulate, is shown to inhibit mammalian DNA polymerase and to suppress inflammation in the mouse ear (Mizushina *et al*., 2013). The potential studies suggest the beneficial effect of rice bran and its bioactive compounds as anti-inflammatory agents.

POTENTIAL APPLICATIONS

Rice bran is used in food as full-fat bran, defatted bran, rice bran oil and protein concentrates (Prakash & Ramaswamy, 1996). The image of rice bran powder is shown in Figure 16.2. Breads developed by addition of 5% and 10% bran have acceptable quality, and bread added with 15% bran is acceptable with bread improver and shelf life increases with increase amount of bran. Therefore, the inexpensive rice bran contributes to food security and can serve as a functional ingredient for high-fibre breads (Fabian & Ju, 2011). Rice bran protein extracts also have promising emulsification properties to be used as natural emulsifier in food products (Lee *et al*., 2004) particularly in the food products such as coffee whiteners, toppings, beverages, confectionary, meat and bakery products (Hamada, 2000) as well as in liquid foods like milk and other drinks (Watchararuji *et al*., 2008).

Figure 16.2 Rice bran powder

Rice bran fractions can be used to produce low-fat and high-fibre bakery products (Kennedy & Burlingame, 2003). For example, rice bran, Torticas, a traditional Cuban bakery product with higher protein, fat, fibre and ash, is produced from parboiled rice bran (Zumbado *et al*., 1997). Rice bran fibre is reported to contain high amounts of functional proteins and fats, along with antioxidants, vitamins and trace minerals in addition to being a concentrated source of fibre as the nutritional and functional ingredients in food products (Fabian & Ju, 2011; Zumbado *et al*., 1997). Stabilised rice bran and its nutrient components have been used as ingredients in various food products such as bread (Hu *et al*., 2009), cookies (Bhanger *et al*., 2008), pizza (de Delahaye *et al*., 2005), beverages (Faccin *et al*., 2009) and tuna oil (Chotimarkorn *et al*., 2008), and can be used to enrich wheat flour as a source of dietary fibre (de Delahaye *et al*., 2005).

Rice bran oil (RBO) has been traditionally consumed and it is known as a 'healthy oil' in the Asian rice producing countries with growing interest in the Western markets (Donald, 2001; Sugano & Tsuji, 1997; Jariwalla, 2000). RBO has been used for frying food due to its oxidative stability and flavour (Mishra & Sharma, 2014). It is now considered as a good substitute for vegetable oils because it contains oryzanol, omega-3 and omega-6 fatty acids as well as non-saponifiable components (Sayre *et al*., 1985; Gopala Krishna, 2002). RBO has an impressive nutritional quality which makes it suitable for nutraceutical products. Production of margarine from RBO gives health benefits with reduced saturated fats and trans-fatty acids. It also has industrial potential, particularly in snack food preparation, due to the great frying stability and with a longer half-life and nutty flavour (Sarkar & Bhattacharyya, 1991).

Rice bran can also be used in anti-ageing products because its oryzanol component can act as a protective agent against UV light-induced lipid peroxidation and as a potent sunscreen agent (Noboru & Yusho, 1970). The oryzanol in rice bran can also stimulate hair growth and prevent skin ageing (Shugo, 1979). Besides, rice bran contains approximately 500 ppm of tocotrienols (Eitenmiller, 1997). When applied the tocotrienol-rich fraction to skin, it can be penetrated and get absorbed rapidly. The efficacy of tocotrienol-containing sunscreens that reduce penetration or absorb ultraviolet radiation is augmented by adding tocotrienol from RBO in the products (Noboru & Yusho, 1970; Shugo, 1979).

Apart from widely used in food industry, rice bran also being used in pharmaceutical industry due to its unique properties, high medicinal value and therapeutics applications (Cicero & Gaddi, 2001). It looks promising for application of rice bran-derived products as functional foods with beneficial health effects and as preventive or clinical medicine. With disease-preventing, therapeutic properties and with no or little toxic effect, rice bran can be formulated for prevention of diseases as well as containing dietary factors with improved nutritive value for the future.

CONCLUSION AND RECOMMENDATIONS

Due to its composition, nutritional profile and functional ingredients, rice bran has many applications in our daily life. As a diet, it is characterised by the high dietary fibre and low saturated fat. To some extent, strong evidence is available for consumption of rice bran that may reduce the risk of cardiovascular diseases and colon cancer. Rice bran is also rich in phytonutrients with remarkable medicinal properties. From studies conducted, products from rice bran are demonstrated to be safe. The bioactive components of rice bran such as inositol and related compounds, phytic acid or phytate, rice bran oil, ferulic acid, γ-oryzanol, plant sterols and tocotrienols and rice-bran products have potential applications as nutritional ingredients in the context of their utility as functional foods.

The components in rice bran demonstrated to have protective effect against several types of disease including cancer, hyperlipidaemia, diabetes, kidney stones and heart disease. To date, rice bran phytonutrients can be potentially used as natural medicines for prevention or treatment of chronic diseases. Considering the rice grain production in the world, it is estimated that high volumes of rice bran by-products will be produced. To adding value to these by-products and producing healthy foods from them, usage of rice bran is essential. Lastly, rice bran is an inexpensive functional food with a range of bioactive compounds.

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